

Muhammad Umer Khan Niazi

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SUMMARY

Second-year AI student at UET Lahore with experience building end-to-end NLP pipelines and deployable ML tools. Focused on practical machine learning, systems programming, and writing production-quality code.

EDUCATION

University of Engineering and Technology <i>BS Artificial Intelligence GPA: 3.25/4.0</i>	Lahore, Pakistan 2025 – Present
Forman Christian College <i>Intermediate Computer Science</i>	Lahore, Pakistan 2022 – 2024

PROJECTS

News NLP Pipeline <i>Python, BERTopic, scikit-learn, spaCy, Transformers, Streamlit</i>	2026 – Present
- Processed 350k+ news articles and headlines from Dawn (2010–2025) through an end-to-end NLP pipeline - Applied BERTopic with UMAP and KMeans for unsupervised topic discovery across 15 years of coverage - Built downstream tasks including sentiment analysis, named entity recognition, and news category classification - Deployed interactive Streamlit dashboard for exploring results, topic distributions, and visualizations	
GB Wildlife Classifier <i>Python, PyTorch, Streamlit</i>	2026
- Trained a ResNet18 transfer learning pipeline to classify images of three endangered Himalayan species: brown bear, markhor, and snow leopard, achieving 98.16% validation accuracy - Froze early layers and trained a custom 3-class fully connected head with augmentation to prevent overfitting on a limited dataset - Stress-tested against out-of-distribution data and documented failure modes where visual and taxonomic similarity causes confident mispredictions - Added an 80% confidence threshold to surface unknown inputs rather than force a classification	
Shelter Outcome Predictor <i>Python, scikit-learn, Pandas, Streamlit</i>	2026
- Trained a Random Forest classifier on 62k+ shelter intake records to predict adoption, transfer, or euthanasia risk - Applied balanced class weighting to prioritize recall on at-risk animals, achieving 63% euthanasia recall - Deployed interactive Streamlit app for per-animal risk prediction with probability breakdown	
WMUS <i>Python, curses</i>	2026 – Present
- Designed modular architecture separating audio playback, UI rendering, and state management - Built a terminal music player for Windows with fuzzy search, vim-style navigation, and configurable keybindings - Packaged with a versioned installer and fast metadata caching for large music libraries	

TECHNICAL SKILLS

Languages: Python, C++, HTML, CSS

Tools & Platforms: Git, Linux, Streamlit, SQLite

Machine Learning & NLP: NumPy, Pandas, Matplotlib, scikit-learn, Seaborn, PyTorch, Transformers, Hugging Face, BERTopic